



Cairo University - Faculty Of
Engineering
EECE Department
Virtual Networks Lab using Linux and
the CORE Network Emulator



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1 Effect of TCP Window Size

1.1 Question 2.i

The Throughput Appears to be increasing with Window Size (as Expected) with Zero Retransmissions up to a certain limit where the receiver Buffer gets Full at 6KB and tries to increase it's buffer size so performance is increases again thus making a zigzag type of graph, then at 24KB the Buffer can't get any larger so Performance is Poor.

$$Throughput = \frac{Data\ Transferred}{Interval\ Time}$$

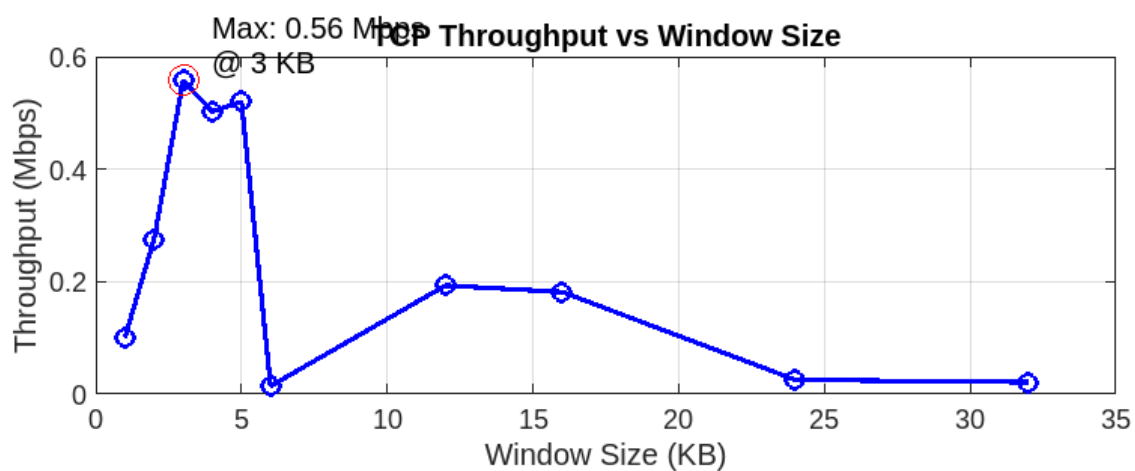


Figure 1: TCP Window Size vs Throughput



Figure 2: TCP Window Size vs Retransmissions

Window Size (KB)	Data (MB)	Throughput (MB/s)	Retransmissions
1	4.01	0.10025	0
2	11.00	0.27500	0
3	22.30	0.55750	0
4	20.10	0.50250	0
5	20.80	0.52000	0
6	0.58	0.01450	345
12	7.73	0.19325	1323
16	7.27	0.18175	1065
24	1.022	0.02555	402
32	0.853	0.021325	404

Table 1: TCP Window Size vs Throughput Data

1.2 Question 2.ii

Frame length is 76 bytes

IPv4 Header length : 20 bytes

TCP header length : 40 bytes

Ethernet = 16 bytes

```
▼ Frame 14025: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
  Arrival Time: Jun  2, 2025 15:00:05.785871000 EEST
  Epoch Time: 1748865605.785871000 seconds
  [Time delta from previous captured frame: 0.000624000 seconds]
  [Time delta from previous displayed frame: 0.000624000 seconds]
  [Time since reference or first frame: 66.350751000 seconds]
  Frame Number: 14025
  Frame Length: 76 bytes (608 bits)
  Capture Length: 76 bytes (608 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: sll:ip:tcp]
  [Coloring Rule Name: TCP SYN/FIN]
  [Coloring Rule String: tcp.flags & 0x02 || tcp.flags.fin == 1]
  ▸ Linux cooked capture
  ▼ Internet Protocol Version 4, Src: 10.0.12.20 (10.0.12.20), Dst: 10.0.13.20 (10.0.13.20)
    Version: 4
    Header length: 20 bytes
    ▸ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
    Total Length: 60
    Identification: 0x741c (29724)
    ▸ Flags: 0x02 (Don't Fragment)
    Fragment offset: 0
    Time to live: 61
    Protocol: TCP (6)
    ▸ Header checksum: 0x9c78 [correct]
    Source: 10.0.12.20 (10.0.12.20)
    Destination: 10.0.13.20 (10.0.13.20)
  ▼ Transmission Control Protocol, Src Port: 60642 (60642), Dst Port: targus-getdata1 (5201), Seq: 0, Len: 0
    Source port: 60642 (60642)
    Destination port: targus-getdata1 (5201)
    [Stream index: 11]
    Sequence number: 0 (relative sequence number)
    Header length: 40 bytes
    ▸ Flags: 0x002 (SYN)
    Window size value: 1116
    [Calculated window size: 1116]
    ▼ Checksum: 0x2d56 [validation disabled]
      [Good Checksum: False]
      [Bad Checksum: False]
    ▼ Options: (20 bytes)
      Maximum segment size: 1460 bytes
      TCP SACK Permitted Option: True
      ▼ Timestamps: TSval 462622, TSecr 0
        Kind: Timestamp (8)
        Length: 10
        Timestamp value: 462622
        Timestamp echo reply: 0
        No-Operation (NOP)
      ▼ Window scale: 0 (multiply by 1)
        Kind: Window Scale (3)
        Length: 3
```

Figure 3: Sent packet from node n7

1.3 Question 2.iii

Frame length is 68 bytes

IPv4 Header length : 20 bytes

TCP header length : 32 bytes

Ethernet = 16 bytes

Ack number = 42 & Sequence No. = 2 (Both Relative)

```

Frame 13993: 68 bytes on wire (544 bits), 68 bytes captured (544 bits)
  Arrival Time: Jun  2, 2025 15:00:05.778537000 EEST
  Epoch Time: 1748865605.778537000 seconds
  [Time delta from previous captured frame: 0.000669000 seconds]
  [Time delta from previous displayed frame: 0.000669000 seconds]
  [Time since reference or first frame: 66.343417000 seconds]
  Frame Number: 13993
  Frame Length: 68 bytes (544 bits)
  Capture Length: 68 bytes (544 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: sll:ip:tcp]
  [Coloring Rule Name: Bad TCP]
  [Coloring Rule String: tcp.analysis.flags]
Linux cooked capture
Internet Protocol Version 4, Src: 10.0.13.20 (10.0.13.20), Dst: 10.0.12.20 (10.0.12.20)
  Version: 4
  Header length: 20 bytes
  Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
  Total Length: 52
  Identification: 0xfecc (65228)
  Flags: 0x02 (Don't Fragment)
  Fragment offset: 0
  Time to live: 61
  Protocol: TCP (6)
  Header checksum: 0x11d0 [correct]
  Source: 10.0.13.20 (10.0.13.20)
  Destination: 10.0.12.20 (10.0.12.20)
Transmission Control Protocol, Src Port: targus-getdata1 (5201), Dst Port: 60641 (60641), Seq: 2, Ack: 42, Len: 0
  Source port: targus-getdata1 (5201)
  Destination port: 60641 (60641)
  [Stream index: 10]
  Sequence number: 2 (relative sequence number)
  Acknowledgement number: 42 (relative ack number)
  Header length: 32 bytes
  Flags: 0x010 (ACK)
  Window size value: 905
  [Calculated window size: 14480]
  [Window size scaling factor: 16]
  Checksum: 0x2d4e [validation disabled]
    [Good Checksum: False]
    [Bad Checksum: False]
  Options: (12 bytes)
    No-Operation (NOP)
    No-Operation (NOP)
    Timestamps: TSval 462621, TSecr 462611
      Kind: Timestamp (8)
      Length: 10
      Timestamp value: 462621
      Timestamp echo reply: 462611
  [SEQ/ACK analysis]
    [TCP Analysis Flags]
```

Figure 4: Received from node n11

2 TCP Short versus Long Paths

2.1 Question 3.i

The Lower Performance is due to larger Delay between n7 & n8 ($4 * 160 + 2 * 110\mu\text{sec}$) compared to n7 & n11 ($3 * 160 + 110\mu\text{sec}$)

Terminal Output:

```
64 bytes from 10.0.11.20: icmp_req=3 ttl=59 time=6.04 ms
^Z
[1]+  Stopped                  ping 10.0.11.20
n7.conf# iperf3 -c 10.0.11.20 -t 40 -i 10 -w 4k
Connecting to host 10.0.11.20, port 5201
[4] local 10.0.12.20 port 48635 connected to 10.0.11.20 port 5201
[ID] Interval          Transfer    Bandwidth    Retr  Cwnd
[4]   0.00-10.00    sec   2.00 MBytes  1.68 Mbits/sec    1   8.48 KBytes
[4]  10.00-20.00    sec   1.71 MBytes  1.44 Mbits/sec    0   8.48 KBytes
[4]  20.00-30.12    sec   1.63 MBytes  1.35 Mbits/sec    0   8.48 KBytes
[4]  30.12-40.00    sec   1.94 MBytes  1.64 Mbits/sec    4   8.48 KBytes
- - - - -
[ID] Interval          Transfer    Bandwidth    Retr
[4]   0.00-40.00    sec   7.28 MBytes  1.53 Mbits/sec    5
[4]   0.00-40.00    sec   7.27 MBytes  1.53 Mbits/sec
iperf Done.
```

Window Size (KB)	Data (MB)	Throughput (MB/s)
4(old)	20.10	0.50250
4(new)	7.28	0.182

Table 2: TCP With Different Propagation Delay

3 Higher Link Capacity with Drops versus Reliable Lower Capacity

Case #	Data (MB)	Throughput (MB/s)	Throughput (Mbps)	Retransmissions
A	10.10	0.2525	2.13	4
B	9.91	0.2478	2.08	2
C	2.09	0.0523	0.438	108
D	0.728	0.0182	0.149	91
E	6.70	0.1675	1.41	56
F	8.74	0.2185	1.83	25

Table 3: 4KB Window Size Performance Variability (40-second tests)

3.1 Question 4.i

Throughput in case A is the best among these results due to high rate and no loss (Highest Capacity)

Also Case B is better Than C as the Bandwidth isn't really an issue with this window Size So Loss from both directions Really Impacts the Performance of Retransmissions and thus the throughput

3.2 Question 4.ii

Case B is better as there's no loss with adequate Capacity For given window Size unlike C which ReTransmits packets

3.3 Question 4.iii

Case E is better Loss in the direction from n4 to n5 is Just Dropped Acks which doesn't effect the throughput that much especially in Go-BACK-N protocol

4 OSPF Link Cost Changes

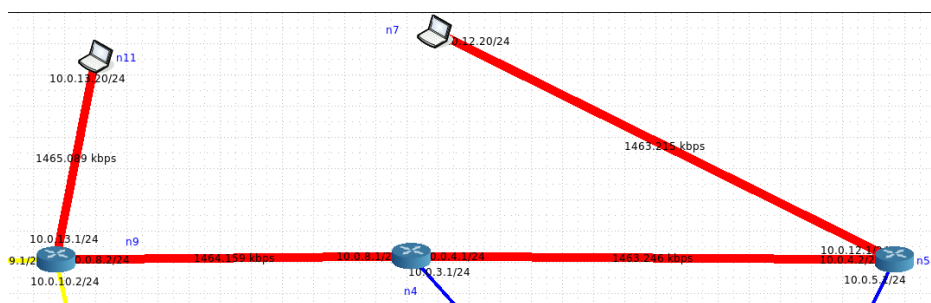


Figure 5: Path between n7 and n11

4.1 Question 5.i

in Figure 6a, when the cost was changed to 40, the path changed from node 4 to node 6 as in Figure:7 the traceroute changed (in graph 6b),also the Throughput Slightly decreased ,Which's due to Cost.

```
root@n7:/tmp/pycore.50997/n7.conf# iperf3 -c 10.0.13.20 -t 500 -i 20 -w 4k
Connecting to host 10.0.13.20, port 5201
[ 4] local 10.0.12.20 port 47308 connected to 10.0.13.20 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4]  0.00-20.00  sec  7.22 MBytes  3.03 Mbits/sec  3    8.48 KBytes
[ 4]  20.00-40.00  sec  4.52 MBytes  1.90 Mbits/sec  3    8.48 KBytes
[ 4]  40.00-60.03  sec  7.85 MBytes  3.29 Mbits/sec  0    8.48 KBytes
[ 4]  60.03-80.00  sec  6.24 MBytes  2.62 Mbits/sec  2    8.48 KBytes
[ 4]  80.00-100.00 sec  6.39 MBytes  2.68 Mbits/sec  1    8.48 KBytes
[ 4]  100.00-120.00 sec  7.47 MBytes  3.13 Mbits/sec  1    8.48 KBytes
[ 4]  120.00-140.05 sec  4.84 MBytes  2.03 Mbits/sec  1    8.48 KBytes
[ 4]  140.05-160.00 sec  3.31 MBytes  1.64 Mbits/sec  1    8.48 KBytes
[ 4]  160.00-180.00 sec  6.79 MBytes  2.85 Mbits/sec  1    8.48 KBytes
[ 4]  180.00-200.00 sec  8.54 MBytes  3.58 Mbits/sec  1    8.48 KBytes
[ 4]  200.00-220.01 sec  5.56 MBytes  2.33 Mbits/sec  2    8.48 KBytes
[ 4]  220.01-240.00 sec  7.20 MBytes  3.02 Mbits/sec  4    8.48 KBytes
[ 4]  240.00-260.00 sec  5.82 MBytes  2.44 Mbits/sec  0    8.48 KBytes
[ 4]  260.00-280.06 sec  6.48 MBytes  2.71 Mbits/sec  0    8.48 KBytes
[ 4]  280.06-300.00 sec  6.71 MBytes  2.82 Mbits/sec  2    8.48 KBytes
[ 4]  300.00-320.00 sec  5.92 MBytes  2.48 Mbits/sec  2    8.48 KBytes
[ 4]  320.00-340.00 sec  6.48 MBytes  2.72 Mbits/sec  2    8.48 KBytes
[ 4]  340.00-360.08 sec  6.62 MBytes  2.77 Mbits/sec  3    2.83 KBytes
[ 4]  360.08-380.01 sec  5.82 MBytes  2.45 Mbits/sec  5    8.48 KBytes
[ 4]  380.01-400.07 sec  6.27 MBytes  2.62 Mbits/sec  3    7.07 KBytes
[ 4]  400.07-420.00 sec  6.62 MBytes  2.79 Mbits/sec  1    8.48 KBytes
[ 4]  420.00-440.00 sec  6.45 MBytes  2.71 Mbits/sec  2    8.48 KBytes
[ 4]  440.00-460.06 sec  5.95 MBytes  2.49 Mbits/sec  2    2.83 KBytes
[ 4]  460.06-480.00 sec  6.98 MBytes  2.94 Mbits/sec  1    8.48 KBytes
[ 4]  480.00-500.02 sec  5.49 MBytes  2.30 Mbits/sec  0    8.48 KBytes

[ ID] Interval      Transfer    Bandwidth  Retr
[ 4]  0.00-500.02 sec  158 MBytes  2.65 Mbits/sec  43      sender
[ 4]  0.00-500.02 sec  158 MBytes  2.65 Mbits/sec              receiver

iperf Done.
root@n7:/tmp/pycore.50997/n7.conf# traceroute 10.0.13.20
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1  10.0.12.1 (10.0.12.1)  0.549 ms  0.462 ms  0.446 ms
 2  10.0.5.2 (10.0.5.2)  2.472 ms  2.703 ms  2.785 ms
 3  * * *
 4  10.0.13.20 (10.0.13.20)  8.013 ms  7.990 ms  7.987 ms
root@n7:/tmp/pycore.50997/n7.conf#
```

(a) Cost Changed to 40

```
root@n7:/tmp/pycore.50997/n7.conf# iperf3 -c 10.0.13.20 -t 500 -i 20 -w 4k
Connecting to host 10.0.13.20, port 5201
[ 4] local 10.0.12.20 port 47306 connected to 10.0.13.20 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4]  0.00-20.00  sec  8.16 MBytes  3.42 Mbits/sec  1    8.48 KBytes
[ 4]  20.00-40.00  sec  8.60 MBytes  3.61 Mbits/sec  0    8.48 KBytes
[ 4]  40.00-60.02  sec  6.87 MBytes  2.88 Mbits/sec  2    8.48 KBytes
[ 4]  60.02-80.00  sec  8.03 MBytes  3.37 Mbits/sec  2    8.48 KBytes
[ 4]  80.00-100.00 sec  5.27 MBytes  2.21 Mbits/sec  4    8.48 KBytes
[ 4]  100.00-120.01 sec  8.44 MBytes  3.54 Mbits/sec  1    8.48 KBytes
[ 4]  120.01-140.00 sec  6.95 MBytes  2.92 Mbits/sec  1    8.48 KBytes
[ 4]  140.00-160.01 sec  5.33 MBytes  2.23 Mbits/sec  0    8.48 KBytes
[ 4]  160.01-180.00 sec  6.46 MBytes  2.71 Mbits/sec  2    8.48 KBytes
[ 4]  180.00-200.00 sec  8.33 MBytes  3.49 Mbits/sec  1    8.48 KBytes
[ 4]  200.00-220.00 sec  8.21 MBytes  3.44 Mbits/sec  0    8.48 KBytes
[ 4]  220.00-240.03 sec  6.53 MBytes  2.73 Mbits/sec  0    8.48 KBytes
[ 4]  240.03-260.00 sec  7.05 MBytes  2.96 Mbits/sec  2    8.48 KBytes
[ 4]  260.00-280.00 sec  7.48 MBytes  3.14 Mbits/sec  1    8.48 KBytes
[ 4]  280.00-300.00 sec  5.45 MBytes  2.29 Mbits/sec  1    8.48 KBytes
[ 4]  300.00-320.00 sec  8.51 MBytes  3.57 Mbits/sec  0    8.48 KBytes
[ 4]  320.00-340.00 sec  4.95 MBytes  2.08 Mbits/sec  2    8.48 KBytes
[ 4]  340.00-360.00 sec  9.45 MBytes  3.96 Mbits/sec  4    8.48 KBytes
[ 4]  360.00-380.00 sec  7.25 MBytes  3.04 Mbits/sec  2    8.48 KBytes
[ 4]  380.00-400.00 sec  11.0 MBytes  4.61 Mbits/sec  0    8.48 KBytes
[ 4]  400.00-420.03 sec  8.92 MBytes  3.73 Mbits/sec  0    8.48 KBytes
[ 4]  420.03-440.04 sec  6.90 MBytes  2.90 Mbits/sec  0    8.48 KBytes
[ 4]  440.04-460.00 sec  6.09 MBytes  2.56 Mbits/sec  2    8.48 KBytes
[ 4]  460.00-480.00 sec  8.16 MBytes  3.42 Mbits/sec  1    8.48 KBytes
[ 4]  480.00-500.00 sec  7.11 MBytes  2.98 Mbits/sec  1    8.48 KBytes

[ ID] Interval      Transfer    Bandwidth  Retr
[ 4]  0.00-500.00 sec  185 MBytes  3.11 Mbits/sec  30      sender
[ 4]  0.00-500.00 sec  185 MBytes  3.11 Mbits/sec              receiver

iperf Done.
root@n7:/tmp/pycore.50997/n7.conf# traceroute 10.0.13.20
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1  * * *
 2  * * *
 3  * * *
 4  10.0.13.20 (10.0.13.20)  4.989 ms  4.978 ms  4.972 ms
root@n7:/tmp/pycore.50997/n7.conf# traceroute 10.0.13.20
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1  10.0.12.1 (10.0.12.1)  0.505 ms  0.457 ms  0.444 ms
 2  10.0.4.1 (10.0.4.1)  0.930 ms  0.936 ms  1.004 ms
 3  * * *
 4  10.0.13.20 (10.0.13.20)  52.519 ms  52.456 ms  52.459 ms
root@n7:/tmp/pycore.50997/n7.conf#
```

(b) Before Changing The Cost

Figure 6: Network performance analysis

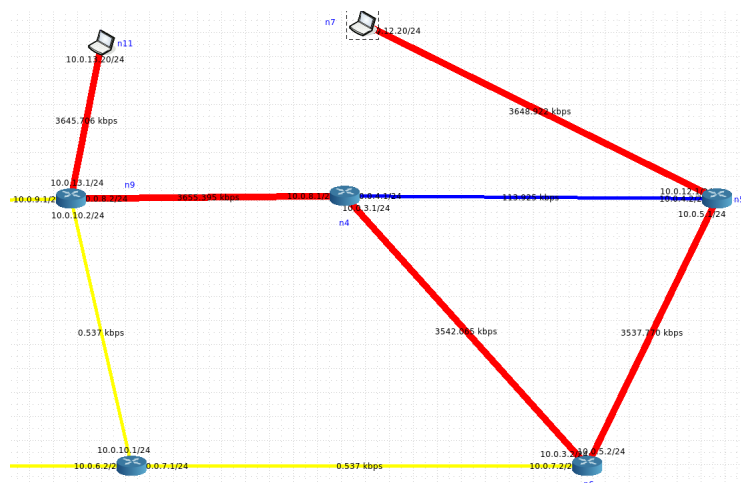
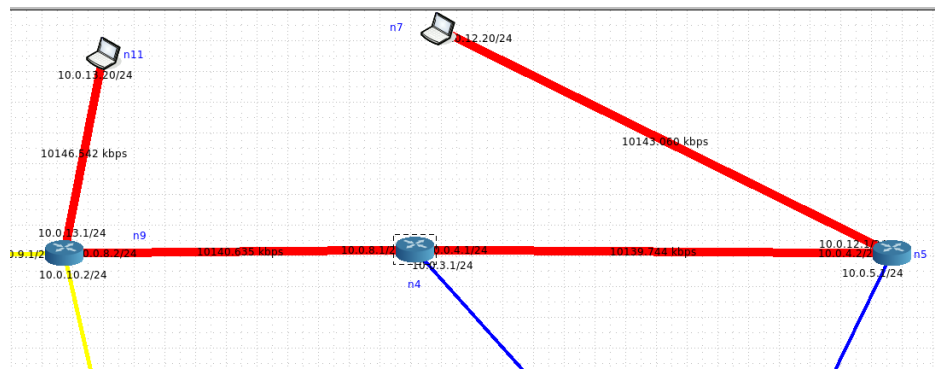


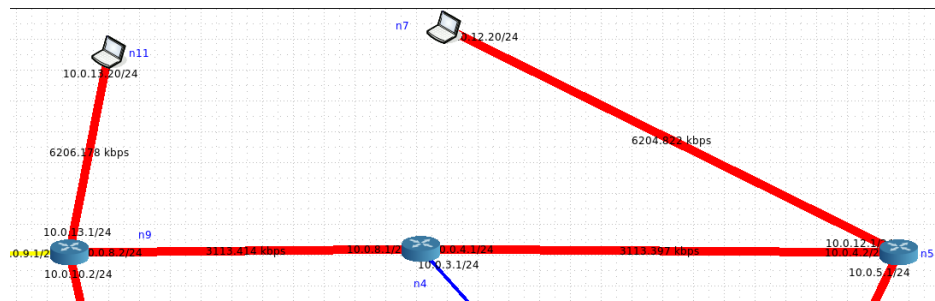
Figure 7: New path after cost change

4.2 Question 5.ii

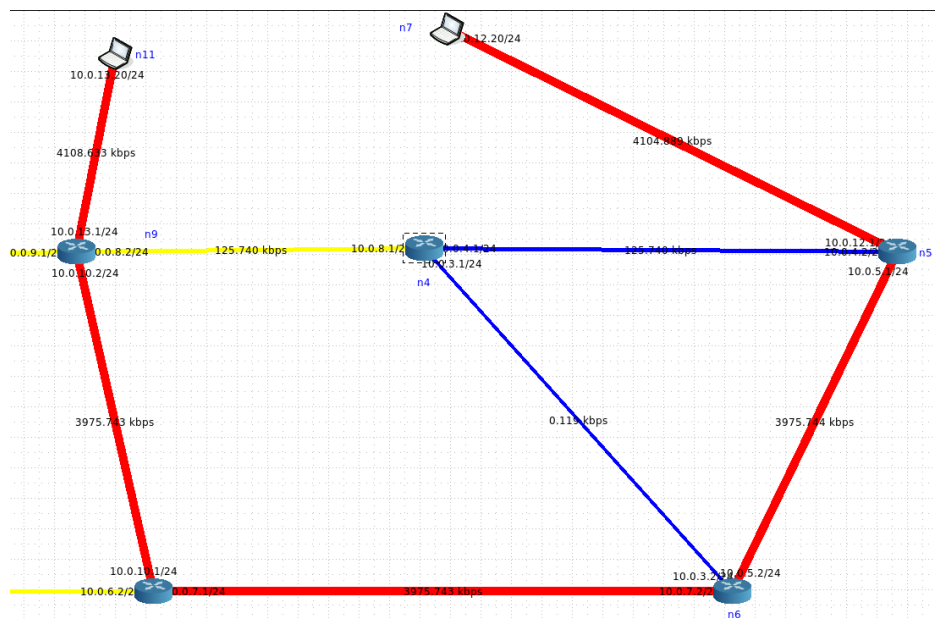
as the cost change from node 4 to node 5 the transfer from node 11 to node 7 decreased Slightly and the bandwidth by changing it's route, while for node 7 to node 11 it doesn't get effected as the cost from node 5 to 4 is 10



(a) Original Path



(b) New Path N7



(c) New Path N11

Figure 8: Network paths before and after cost changes

```

root@n7:/tmp/pycore.50937/n7.conf# iperf3 -c 10.0.13.20 -t 500 -i 20 -w 4k
Connecting to host 10.0.13.20, port 5201
[ 4] local 10.0.12.20 port 47312 connected to 10.0.13.20 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4] 0.00-20.00 sec  3.33 MBytes 3.91 Mbits/sec  0    14.1 KBytes
[ 4] 20.00-40.00 sec  6.80 MBytes 2.84 Mbits/sec  2    8.48 KBytes
[ 4] 40.00-60.00 sec  8.75 MBytes 3.68 Mbits/sec  0    8.48 KBytes
[ 4] 60.00-80.00 sec  5.96 MBytes 2.50 Mbits/sec  0    8.48 KBytes
[ 4] 80.00-100.00 sec  8.12 MBytes 3.41 Mbits/sec  1    8.48 KBytes
[ 4] 100.00-120.00 sec  9.20 MBytes 3.44 Mbits/sec  4    8.48 KBytes
[ 4] 120.00-140.00 sec  7.36 MBytes 3.09 Mbits/sec  1    8.48 KBytes
[ 4] 140.00-160.00 sec  7.59 MBytes 3.18 Mbits/sec  0    8.48 KBytes
[ 4] 160.00-180.00 sec  7.44 MBytes 3.12 Mbits/sec  1    8.48 KBytes
[ 4] 180.00-200.00 sec  10.3 MBytes 4.31 Mbits/sec  1    8.48 KBytes
[ 4] 200.00-220.00 sec  6.70 MBytes 2.81 Mbits/sec  1    8.48 KBytes
[ 4] 220.00-240.00 sec  6.17 MBytes 2.59 Mbits/sec  3    8.48 KBytes
[ 4] 240.00-260.00 sec  8.02 MBytes 3.37 Mbits/sec  1    8.48 KBytes
[ 4] 260.00-280.00 sec  10.4 MBytes 4.35 Mbits/sec  2    8.48 KBytes
[ 4] 280.00-300.00 sec  8.61 MBytes 3.61 Mbits/sec  1    8.48 KBytes
[ 4] 300.00-320.00 sec  7.61 MBytes 3.17 Mbits/sec  5    4.24 KBytes
[ 4] 320.00-340.00 sec  6.98 MBytes 2.95 Mbits/sec  2    8.48 KBytes
[ 4] 340.00-360.00 sec  9.45 MBytes 3.97 Mbits/sec  4    8.48 KBytes
[ 4] 360.00-380.00 sec  8.13 MBytes 3.41 Mbits/sec  1    8.48 KBytes
[ 4] 380.00-400.00 sec  7.44 MBytes 3.12 Mbits/sec  2    8.48 KBytes
[ 4] 400.00-420.00 sec  8.23 MBytes 3.44 Mbits/sec  4    8.48 KBytes
[ 4] 420.00-440.00 sec  6.42 MBytes 2.70 Mbits/sec  2    8.48 KBytes
[ 4] 440.00-460.00 sec  8.56 MBytes 3.59 Mbits/sec  0    8.48 KBytes
[ 4] 460.00-480.00 sec  6.58 MBytes 2.80 Mbits/sec  2    8.48 KBytes
[ 4] 480.00-500.00 sec  7.36 MBytes 3.09 Mbits/sec  3    8.48 KBytes
--
[ ID] Interval      Transfer    Bandwidth  Retr
[ 4] 0.00-500.00 sec  197 MBytes 3.30 Mbits/sec  43
[ 4] 0.00-500.00 sec  197 MBytes 3.30 Mbits/sec
sender
receiver

iperf Done.
root@n7:/tmp/pycore.50937/n7.conf#

```

(a) Node 7 to Node 11 (cost=10)

```

root@n11:/tmp/pycore.50937/n11.conf# iperf3 -c 10.0.12.20 -t 500 -i 20 -w 4k
Connecting to host 10.0.12.20, port 5201
[ 4] local 10.0.13.20 port 36198 connected to 10.0.12.20 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4] 0.00-20.00 sec  8.44 MBytes 3.54 Mbits/sec  1    8.48 KBytes
[ 4] 20.00-40.00 sec  8.66 MBytes 3.63 Mbits/sec  0    8.48 KBytes
[ 4] 40.00-60.00 sec  8.23 MBytes 3.45 Mbits/sec  1    8.48 KBytes
[ 4] 60.00-80.00 sec  8.15 MBytes 3.42 Mbits/sec  1    8.48 KBytes
[ 4] 80.00-100.00 sec  6.23 MBytes 2.61 Mbits/sec  0    8.48 KBytes
[ 4] 100.00-120.00 sec  9.27 MBytes 3.47 Mbits/sec  2    8.48 KBytes
[ 4] 120.00-140.00 sec  8.35 MBytes 3.50 Mbits/sec  1    8.48 KBytes
[ 4] 140.00-160.00 sec  7.03 MBytes 2.95 Mbits/sec  2    8.48 KBytes
[ 4] 160.00-180.00 sec  8.03 MBytes 3.37 Mbits/sec  1    8.48 KBytes
[ 4] 180.00-200.00 sec  7.67 MBytes 3.22 Mbits/sec  0    8.48 KBytes
[ 4] 200.00-220.00 sec  9.10 MBytes 3.82 Mbits/sec  2    8.48 KBytes
[ 4] 220.00-240.00 sec  7.89 MBytes 3.31 Mbits/sec  0    8.48 KBytes
[ 4] 240.00-260.00 sec  5.74 MBytes 2.41 Mbits/sec  3    8.48 KBytes
[ 4] 260.00-280.00 sec  8.36 MBytes 3.51 Mbits/sec  0    8.48 KBytes
[ 4] 280.00-300.00 sec  9.17 MBytes 3.84 Mbits/sec  3    8.48 KBytes
[ 4] 300.00-320.00 sec  9.64 MBytes 4.05 Mbits/sec  2    8.48 KBytes
[ 4] 320.00-340.00 sec  7.06 MBytes 2.96 Mbits/sec  3    8.48 KBytes
[ 4] 340.00-360.00 sec  7.80 MBytes 3.27 Mbits/sec  4    8.48 KBytes
[ 4] 360.00-380.00 sec  9.38 MBytes 3.94 Mbits/sec  3    8.48 KBytes
[ 4] 380.00-400.00 sec  7.67 MBytes 3.21 Mbits/sec  2    8.48 KBytes
[ 4] 400.00-420.00 sec  7.52 MBytes 3.15 Mbits/sec  2    8.48 KBytes
[ 4] 420.00-440.00 sec  9.01 MBytes 3.77 Mbits/sec  3    8.48 KBytes
[ 4] 440.00-460.00 sec  6.50 MBytes 2.73 Mbits/sec  1    8.48 KBytes
[ 4] 460.00-480.00 sec  8.14 MBytes 3.41 Mbits/sec  0    8.48 KBytes
[ 4] 480.00-500.00 sec  7.13 MBytes 2.99 Mbits/sec  2    8.48 KBytes
--
[ ID] Interval      Transfer    Bandwidth  Retr
[ 4] 0.00-500.00 sec  199 MBytes 3.34 Mbits/sec  39
[ 4] 0.00-500.00 sec  199 MBytes 3.34 Mbits/sec
sender
receiver

iperf Done.
root@n11:/tmp/pycore.50937/n11.conf#

```

(b) Node 11 to Node 7 (cost=10)

```

root@n7:/tmp/pycore.50937/n7.conf# iperf3 -c 10.0.13.20 -t 500 -i 20 -w 4k
Connecting to host 10.0.13.20, port 5201
[ 4] local 10.0.12.20 port 47316 connected to 10.0.13.20 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4] 0.00-20.00 sec  6.21 MBytes 2.60 Mbits/sec  1    8.48 KBytes
[ 4] 20.00-40.00 sec  7.01 MBytes 2.94 Mbits/sec  2    8.48 KBytes
[ 4] 40.00-60.00 sec  5.68 MBytes 2.38 Mbits/sec  2    8.48 KBytes
[ 4] 60.00-80.00 sec  8.01 MBytes 3.36 Mbits/sec  4    8.48 KBytes
[ 4] 80.00-100.00 sec  6.98 MBytes 2.92 Mbits/sec  4    8.48 KBytes
[ 4] 100.00-120.00 sec  8.98 MBytes 3.77 Mbits/sec  0    8.48 KBytes
[ 4] 120.00-140.00 sec  7.81 MBytes 3.28 Mbits/sec  1    8.48 KBytes
[ 4] 140.00-160.00 sec  7.45 MBytes 3.12 Mbits/sec  2    8.48 KBytes
[ 4] 160.00-180.00 sec  6.84 MBytes 2.87 Mbits/sec  5    8.48 KBytes
[ 4] 180.00-200.00 sec  6.11 MBytes 2.56 Mbits/sec  3    8.48 KBytes
[ 4] 200.00-220.00 sec  4.62 MBytes 1.94 Mbits/sec  10   8.48 KBytes
[ 4] 220.00-240.00 sec  7.66 MBytes 3.21 Mbits/sec  4    8.48 KBytes
[ 4] 240.00-260.00 sec  7.31 MBytes 3.06 Mbits/sec  1    8.48 KBytes
[ 4] 260.00-280.00 sec  8.01 MBytes 3.36 Mbits/sec  0    8.48 KBytes
[ 4] 280.00-300.00 sec  7.11 MBytes 2.98 Mbits/sec  0    8.48 KBytes
[ 4] 300.00-320.00 sec  7.19 MBytes 3.02 Mbits/sec  6    8.48 KBytes
[ 4] 320.00-340.00 sec  9.22 MBytes 3.87 Mbits/sec  4    8.48 KBytes
[ 4] 340.00-360.00 sec  6.24 MBytes 2.62 Mbits/sec  3    8.48 KBytes
[ 4] 360.00-380.00 sec  6.67 MBytes 2.80 Mbits/sec  1    8.48 KBytes
[ 4] 380.00-400.00 sec  8.68 MBytes 3.64 Mbits/sec  0    8.48 KBytes
[ 4] 400.00-420.00 sec  8.18 MBytes 3.43 Mbits/sec  1    8.48 KBytes
[ 4] 420.00-440.00 sec  7.85 MBytes 3.29 Mbits/sec  2    8.48 KBytes
[ 4] 440.00-460.00 sec  6.96 MBytes 2.92 Mbits/sec  1    8.48 KBytes
[ 4] 460.00-480.00 sec  5.85 MBytes 2.45 Mbits/sec  2    8.48 KBytes
[ 4] 480.00-500.00 sec  5.35 MBytes 2.25 Mbits/sec  3    8.48 KBytes
--
[ ID] Interval      Transfer    Bandwidth  Retr
[ 4] 0.00-500.00 sec  178 MBytes 2.93 Mbits/sec  65
[ 4] 0.00-500.00 sec  178 MBytes 2.93 Mbits/sec
sender
receiver

iperf Done.
root@n7:/tmp/pycore.50937/n7.conf# traceroute 10.0.13.20 -q 10 -w 10
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1 10.0.12.1 (10.0.12.1) 0.500 ms 0.434 ms 0.419 ms 0.400 ms 0.385 ms 0.3
68 ms *****
 2 ***** 10.0.3.1 (10.0.3.1) 26.746 ms 26.733 ms 26.799 ms
 3 10.0.8.2 (10.0.8.2) 27.242 ms 27.246 ms *****
 4 10.0.13.20 (10.0.13.20) 2.062 ms 2.112 ms 2.081 ms 2.194 ms 2.057 ms 2
.610 ms *****
root@n7:/tmp/pycore.50937/n7.conf#

```

(c) Node 7 to Node 11 (cost=40)

```

iperf Done.
root@n11:/tmp/pycore.50937/n11.conf# iperf3 -c 10.0.12.20 -t 500 -i 20 -w 4k
Connecting to host 10.0.12.20, port 5201
[ 4] local 10.0.13.20 port 36202 connected to 10.0.12.20 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4] 0.00-20.00 sec  7.79 MBytes 3.27 Mbits/sec  0    14.1 KBytes
[ 4] 20.00-40.00 sec  8.44 MBytes 3.54 Mbits/sec  3    8.48 KBytes
[ 4] 40.00-60.00 sec  6.96 MBytes 2.92 Mbits/sec  2    8.48 KBytes
[ 4] 60.00-80.00 sec  6.44 MBytes 2.70 Mbits/sec  2    8.48 KBytes
[ 4] 80.00-100.00 sec  6.56 MBytes 2.75 Mbits/sec  5    8.48 KBytes
[ 4] 100.00-120.00 sec  7.96 MBytes 3.17 Mbits/sec  1    8.48 KBytes
[ 4] 120.00-140.00 sec  6.39 MBytes 2.68 Mbits/sec  3    8.48 KBytes
[ 4] 140.00-160.00 sec  9.18 MBytes 3.85 Mbits/sec  1    8.48 KBytes
[ 4] 160.00-180.00 sec  7.70 MBytes 3.23 Mbits/sec  1    8.48 KBytes
[ 4] 180.00-200.00 sec  7.84 MBytes 3.28 Mbits/sec  1    8.48 KBytes
[ 4] 200.00-220.00 sec  6.87 MBytes 2.88 Mbits/sec  5    8.48 KBytes
[ 4] 220.00-240.00 sec  6.19 MBytes 2.60 Mbits/sec  1    8.48 KBytes
[ 4] 240.00-260.00 sec  5.42 MBytes 2.27 Mbits/sec  9    8.48 KBytes
[ 4] 260.00-280.00 sec  6.67 MBytes 2.80 Mbits/sec  3    8.48 KBytes
[ 4] 280.00-300.00 sec  7.50 MBytes 3.14 Mbits/sec  1    8.48 KBytes
[ 4] 300.00-320.00 sec  8.82 MBytes 3.70 Mbits/sec  0    8.48 KBytes
[ 4] 320.00-340.00 sec  6.41 MBytes 2.68 Mbits/sec  3    8.48 KBytes
[ 4] 340.00-360.00 sec  8.01 MBytes 3.37 Mbits/sec  2    8.48 KBytes
[ 4] 360.00-380.00 sec  8.07 MBytes 3.38 Mbits/sec  7    8.48 KBytes
[ 4] 380.00-400.00 sec  7.31 MBytes 3.07 Mbits/sec  3    8.48 KBytes
[ 4] 400.00-420.00 sec  5.47 MBytes 2.29 Mbits/sec  3    8.48 KBytes
[ 4] 420.00-440.00 sec  9.78 MBytes 4.10 Mbits/sec  2    8.48 KBytes
[ 4] 440.00-460.00 sec  7.77 MBytes 3.26 Mbits/sec  0    8.48 KBytes
[ 4] 460.00-480.00 sec  7.08 MBytes 2.97 Mbits/sec  2    8.48 KBytes
[ 4] 480.00-500.00 sec  8.53 MBytes 3.58 Mbits/sec  2    8.48 KBytes
--
[ ID] Interval      Transfer    Bandwidth  Retr
[ 4] 0.00-500.00 sec  185 MBytes 3.10 Mbits/sec  62
[ 4] 0.00-500.00 sec  185 MBytes 3.10 Mbits/sec
sender
receiver

iperf Done.
root@n11:/tmp/pycore.50937/n11.conf# traceroute 10.0.12.20 -q 10 -w 10
traceroute to 10.0.12.20 (10.0.12.20), 30 hops max, 60 byte packets
 1 10.0.13.1 (10.0.13.1) 0.513 ms 0.454 ms 0.431 ms 0.415 ms 0.397 ms 0.3
80 ms *****
 2 ***** 10.0.10.1 (10.0.10.1) 0.844 ms 0.795 ms 0.776 ms 0.999 ms
 3 10.0.3.2 (10.0.3.2) 1.525 ms 1.342 ms 1.318 ms 1.251 ms 1.245 ms 1.267
ms 1.269 ms 1.359 ms *****
 4 ***** 10.0.4.2 (10.0.4.2) 2.413 ms 2.374 ms 2.353 ms 2.331 ms 2.336 m
s 2.347 ms
 5 10.0.12.20 (10.0.12.20) 2.845 ms 2.156 ms 12.627 ms 15.728 ms 2.088 ms
2.076 ms 2.236 ms 2.180 ms 2.233 ms 2.240 ms
root@n11:/tmp/pycore.50937/n11.conf#

```

(d) Node 11 to Node 7 (cost=40)

Figure 9: Network performance before and after changing the interface cost

5 OSPF Database Updates

5.1 Question 6.i

as destination have the same cost and ospf support equal cost multiplath so there are more than path has the same cost. So, switch can select any one of them. Also, if one of the paths is down, there should be another one.

```
Hello, this is Quagga (version 0.99.20.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

n2# config terminal
n2(config)# show ip ospf database
% Unknown command.
n2(config)# do show ip ospf database

    OSPF Router with ID (10.0.1.2)

      Router Link States (Area 0.0.0.0)

Link ID      ADV Router   Age  Seq#       CkSum Link
10.0.0.1     10.0.0.1     1001 0x8000000f 0xe8c2 3
10.0.0.2     10.0.0.2     1081 0x80000010 0x8e0e 3
10.0.1.2     10.0.1.2     1050 0x8000000f 0x830f 3
10.0.3.1     10.0.3.1     1554 0x80000011 0x3931 3
10.0.3.2     10.0.3.2     1193 0x80000010 0xa1e4 3
10.0.5.1     10.0.5.1     1054 0x8000000e 0x880b 3
10.0.6.2     10.0.6.2     1160 0x8000000e 0xc2b0 3
10.0.8.2     10.0.8.2     1051 0x80000010 0x1a19 4

      Net Link States (Area 0.0.0.0)

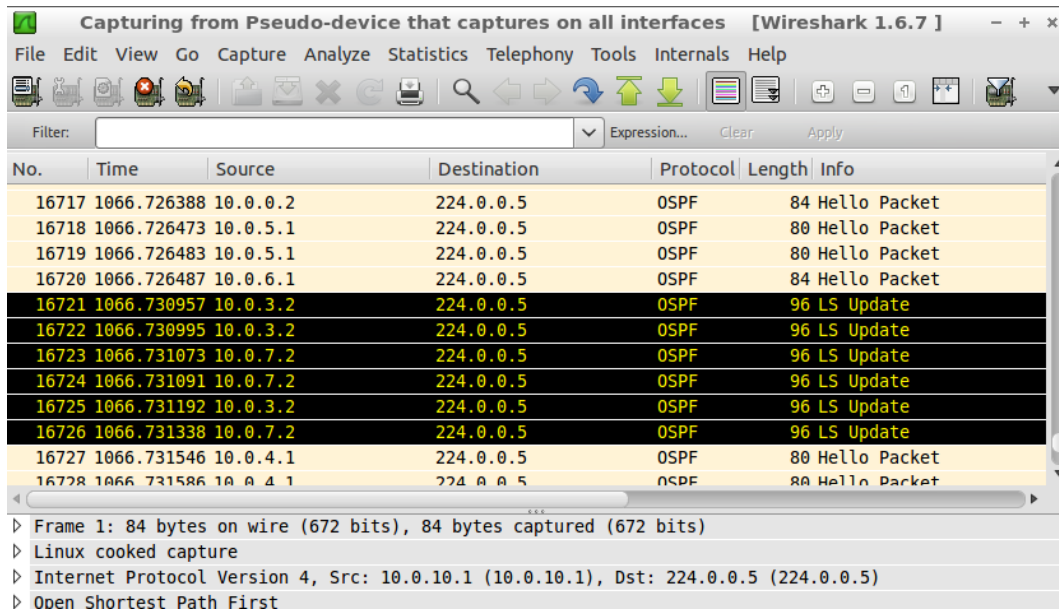
Link ID      ADV Router   Age  Seq#       CkSum
10.0.0.2     10.0.0.2     1011 0x80000006 0x55d0
10.0.1.2     10.0.1.2     940  0x80000006 0x5ac7
10.0.2.1     10.0.1.2     70   0x80000007 0x45dc
10.0.3.2     10.0.3.2     813  0x80000006 0x61b8
10.0.4.2     10.0.5.1     1214 0x80000006 0x5abc

...skipping one line
10.0.6.2     10.0.6.2     1780 0x80000006 0x41d1
10.0.7.1     10.0.6.2     950  0x80000006 0x4fc0
10.0.8.2     10.0.8.2     1631 0x80000006 0x48c2
10.0.9.1     10.0.8.2     1011 0x80000006 0x23e8
10.0.10.2    10.0.8.2     1011 0x80000006 0x57ad

n2(config)# show ip ospf route
% Unknown command.
n2(config)# do show ip ospf route
===== OSPF network routing table =====
N   10.0.0.0/24      [20] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.2.2, eth1
N   10.0.1.0/24      [10] area: 0.0.0.0
                        directly attached to eth0
N   10.0.2.0/24      [10] area: 0.0.0.0
                        directly attached to eth1
N   10.0.3.0/24      [30] area: 0.0.0.0
                        via 10.0.9.1, eth2
N   10.0.4.0/24      [50] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N   10.0.5.0/24      [40] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N   10.0.6.0/24      [20] area: 0.0.0.0
                        via 10.0.1.1, eth0
N   10.0.7.0/24      [30] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N   10.0.8.0/24      [20] area: 0.0.0.0
                        via 10.0.9.1, eth2
--More--
```

Figure 10: OSPF Routing Table Router n2

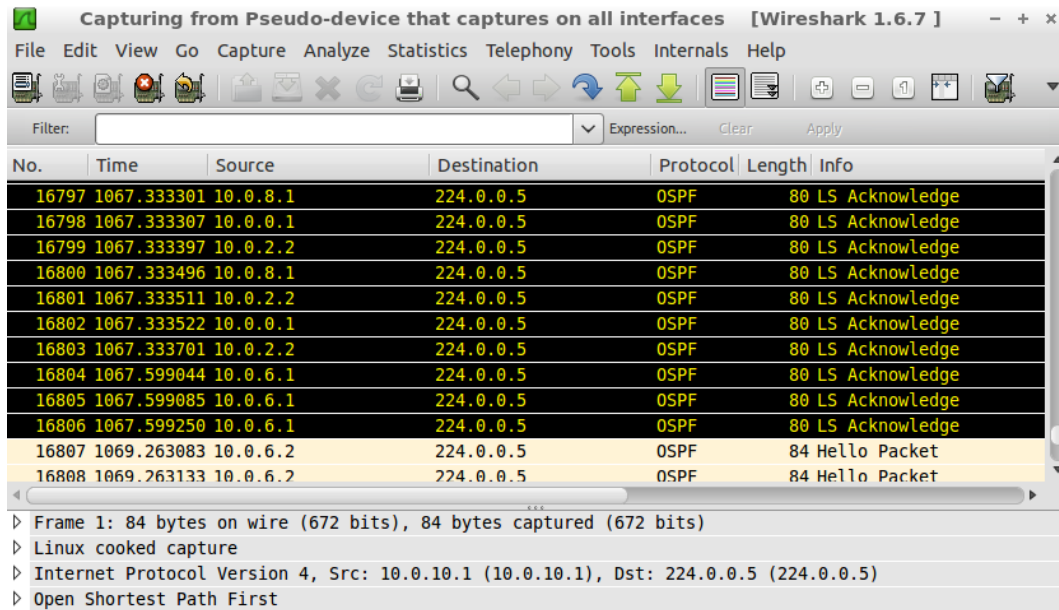
5.2 Question 6.ii



No.	Time	Source	Destination	Protocol	Length	Info
16717	1066.726388	10.0.0.2	224.0.0.5	OSPF	84	Hello Packet
16718	1066.726473	10.0.5.1	224.0.0.5	OSPF	80	Hello Packet
16719	1066.726483	10.0.5.1	224.0.0.5	OSPF	80	Hello Packet
16720	1066.726487	10.0.6.1	224.0.0.5	OSPF	84	Hello Packet
16721	1066.730957	10.0.3.2	224.0.0.5	OSPF	96	LS Update
16722	1066.730995	10.0.3.2	224.0.0.5	OSPF	96	LS Update
16723	1066.731073	10.0.7.2	224.0.0.5	OSPF	96	LS Update
16724	1066.731091	10.0.7.2	224.0.0.5	OSPF	96	LS Update
16725	1066.731192	10.0.3.2	224.0.0.5	OSPF	96	LS Update
16726	1066.731338	10.0.7.2	224.0.0.5	OSPF	96	LS Update
16727	1066.731546	10.0.4.1	224.0.0.5	OSPF	80	Hello Packet
16728	1066.731586	10.0.4.1	224.0.0.5	OSPF	80	Hello Packet

Frame 1: 84 bytes on wire (672 bits), 84 bytes captured (672 bits)
 Linux cooked capture
 Internet Protocol Version 4, Src: 10.0.10.1 (10.0.10.1), Dst: 224.0.0.5 (224.0.0.5)
 Open Shortest Path First

(a) First OSPF Update Message



No.	Time	Source	Destination	Protocol	Length	Info
16797	1067.333301	10.0.8.1	224.0.0.5	OSPF	80	LS Acknowledge
16798	1067.333307	10.0.0.1	224.0.0.5	OSPF	80	LS Acknowledge
16799	1067.333397	10.0.2.2	224.0.0.5	OSPF	80	LS Acknowledge
16800	1067.333496	10.0.8.1	224.0.0.5	OSPF	80	LS Acknowledge
16801	1067.333511	10.0.2.2	224.0.0.5	OSPF	80	LS Acknowledge
16802	1067.333522	10.0.0.1	224.0.0.5	OSPF	80	LS Acknowledge
16803	1067.333701	10.0.2.2	224.0.0.5	OSPF	80	LS Acknowledge
16804	1067.599044	10.0.6.1	224.0.0.5	OSPF	80	LS Acknowledge
16805	1067.599085	10.0.6.1	224.0.0.5	OSPF	80	LS Acknowledge
16806	1067.599250	10.0.6.1	224.0.0.5	OSPF	80	LS Acknowledge
16807	1069.263083	10.0.6.2	224.0.0.5	OSPF	84	Hello Packet
16808	1069.263133	10.0.6.2	224.0.0.5	OSPF	84	Hello Packet

Frame 1: 84 bytes on wire (672 bits), 84 bytes captured (672 bits)
 Linux cooked capture
 Internet Protocol Version 4, Src: 10.0.10.1 (10.0.10.1), Dst: 224.0.0.5 (224.0.0.5)
 Open Shortest Path First

(b) Last ACK Message

Figure 11: OSPF Link State Exchange Timing Analysis

$$time = t1 - t2 = 1,067.599250 - 1,066.730957 = 0.868293sec$$

5.3 Question 6.iii

When we disconnect the N4 node, there should be another node to take its place

```
Hello, this is Quagga (version 0.99.20.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

n2# config terminal
VTY configuration is locked by other VTY
n2# config terminal
n2(config)# do show ip ospf database

      OSPF Router with ID (10.0.1.2)

                Router Link States (Area 0.0.0.0)

Link ID      ADV Router    Age Seq#       CkSum Link
10.0.0.1     10.0.0.1       21 0x80000010 0xe6c3 3
10.0.0.2     10.0.0.2       161 0x80000011 0x8c0f 3
10.0.1.2     10.0.1.2       140 0x80000010 0x8110 3
10.0.3.1     10.0.3.1       81 0x80000016 0xaa47 1
10.0.3.2     10.0.3.2       63 0x80000012 0xe6ad 3
10.0.5.1     10.0.5.1       44 0x80000010 0xf3ae 3
10.0.6.2     10.0.6.2      179 0x8000000f 0xc0b1 3
10.0.8.2     10.0.8.2       40 0x80000012 0x1e28 4

                Net Link States (Area 0.0.0.0)

Link ID      ADV Router    Age Seq#       CkSum
10.0.0.2     10.0.0.2      141 0x80000007 0x53d1
10.0.1.2     10.0.1.2     1820 0x80000006 0x5ac7
10.0.2.1     10.0.1.2      950 0x80000007 0x45dc
10.0.5.1     10.0.5.1     1743 0x80000006 0x5fb6
10.0.6.2     10.0.6.2      840 0x80000007 0x3fd2
10.0.7.1     10.0.6.2       9 0x80000007 0x4dc1
10.0.9.1     10.0.8.2      61 0x80000007 0x21e9
10.0.10.2    10.0.8.2     111 0x80000007 0x55ae

n2(config)# do show ospf route
% Unknown command.
n2(config)# do show ip ospf route
===== OSPF network routing table =====
N    10.0.0.0/24    [20] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.2.2, eth1
N    10.0.1.0/24    [10] area: 0.0.0.0
                        directly attached to eth0
N    10.0.2.0/24    [10] area: 0.0.0.0
                        directly attached to eth1
N    10.0.3.0/24    [40] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N    10.0.4.0/24    [50] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N    10.0.5.0/24    [40] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N    10.0.6.0/24    [20] area: 0.0.0.0
                        via 10.0.1.1, eth0
N    10.0.7.0/24    [30] area: 0.0.0.0
                        via 10.0.1.1, eth0
                        via 10.0.9.1, eth2
N    10.0.8.0/24    [20] area: 0.0.0.0
```

Figure 12: New Routing Table